
epiinfo™
Guide Series
Epi Info™ Guide to StatCalc

EPI INFO™ 7

STAT CALC: PERFORMING STATISTICAL CALCULATIONS

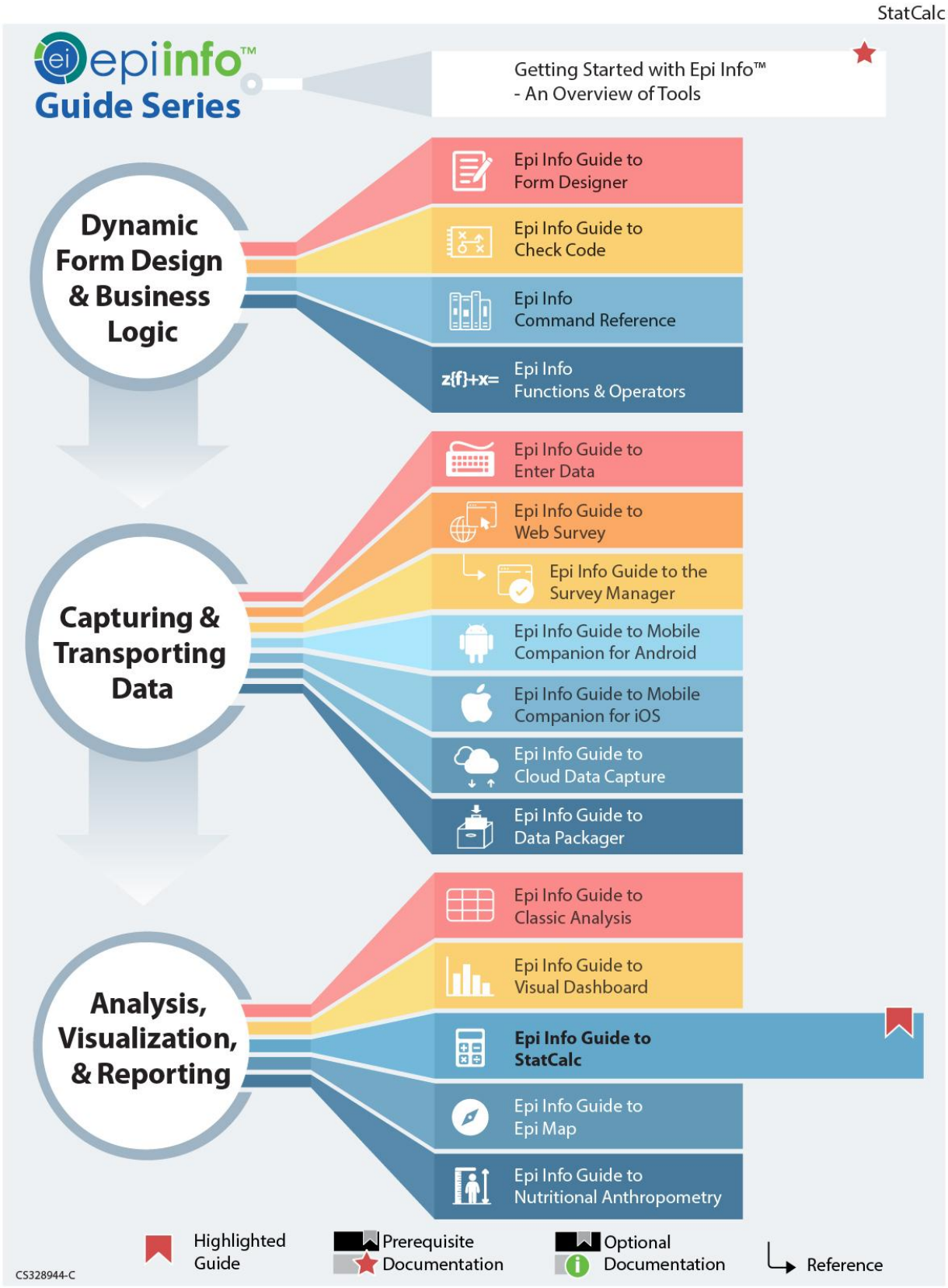
Centers for Disease Control and Prevention

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Preface

Epi Info™ 7 is a public domain suite of interoperable software tools designed for the global community of public health practitioners and researchers. Perform data analysis with epidemiologic statistics, maps, and graphs. Build data entry forms, construct a database, and customize statistics applications. Physicians, epidemiologists, and public health officials without a tech background can easily work with critical data using Epi Info™ tools. Epi Info™ 7 is free of charge. You can Download Epi Info™ 7 from the Centers for Disease Control and Prevention (CDC) website. There are many resources, components, and utilities in the Epi Info™ Suite of Tools. Each major tool has its own guide to help you make the best use of it. The graphic on the opposite page shows the full Epi Info™ Guide Series and the interrelationships between the guides. Each guide in the series has a similar graphic. The current guide is highlighted with a red bookmark icon (). This guide, ***Epi Info™ - StatCalc: Performing Statistical Calculations,*** contains a high-level description of each tool. Use the stand-alone guides for each Epi Info™ tool independently, or in conjunction with related guides for greater functionality. In some cases, related guides are important prerequisites to using the tool discussed in the current guide. The prerequisite documentation has a red star icon (). In other cases, you may find optional guides informative, and the optional documentation is shown with a green information icon (). For detailed instructions on using any of the Epi Info™ tools, refer to its specific guide in the Epi Info™ Guide Series.

The image below shows the interrelationships among the guides in the series. This guide is *Epi Map*.



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Stat Calc: Performing Statistical Calculations

INTRODUCTION

StatCalc is an epidemiologic calculator that produces statistics from summary data. It offers three types of calculations:

1. **Statistics from tables**: Analyzes 2-by-2 to 2-by-9 tables (both single and stratified) to produce:
    - Odds ratios and risk ratios with confidence limits
    - Chi square tests
    - Fisher exact tests
    - Mantel-Haenszel summary odds ratios and chi squares
    - Associated p-values
2. **Sample size and power calculations** for:
    - Population Survey
    - Cohort or Cross-Sectional studies
    - Unmatched Case-Control Study
3. **Chi-square for trend**: Uses the Mantel extension of the Mantel-Haenszel method to test for trends in dose response or case control studies with increasing or decreasing exposures.

How to Use StatCalc

Opening StatCalc

1. From the Epi Info™ 7 main menu, select **StatCalc**.
2. Select one of these calculations using the arrow keys or click:
    - Sample Size & Power
    - Chi Square for Trend
    - Tables (2X2, 2Xn)
    - Poisson (rare event vs. std.)

- Binomial (proportion vs. std.)

You can also select calculations by pressing the keyboard key that matches the highlighted letter on the menu.

3. If you select Sample size and power calculation, you'll see these options:
    - Population Survey
    - Cohort or Cross-Sectional
    - Unmatched Case-Control
4. Use the tab key or click to move between data entry cells.
5. Enter data for your calculation. Results appear automatically as you enter data.
6. Use the Tab key or click on the cell To change values already entered. Type the new information.
7. Print results by right-clicking and then select **Print**.
8. Right-click and select **Save as Image** to save a screenshot of your results.

Analysis of Single and Stratified Tables

These tables work like those in Classic Analysis. You can analyze both single and stratified 2-by-2 tables to calculate odds ratios, risk ratios, various chi square tests, Fisher exact tests, and Mantel-Haenszel statistics.

Assumptions

- Cell values must be correspond ~~correspond~~ to counts representing records that meet the specifications in the margin and stratum labels.
- Study types:
    - **Case-control study**: Ill and well individuals are selected first, then exposure status is determined.
    - **Cross-sectional study**: Neither illness nor exposure is predetermined; both are identified during the study.

- **Cohort study**: Exposed and unexposed individuals are selected first, then illness status is determined.

In cohort studies, you can calculate the relative risk directly. In case-control studies, the odds ratio approximates the relative risk only if the disease is rare (less than 1 case per 20 individuals).

For valid results, outcomes in each record must be independent of other records. Remove confounding by stratifying on confounding variables.

Single 2-by-2 Tables

Two-by-two tables help explore associations between exposure to risk factors and disease. The StatCalc table shows Exposure on the left and Disease across the top.

When analyzing yes-no questions about disease and exposure risk, StatCalc produces statistics that test to determine relationships between them. An association is suggested by an odds ratio or relative risk that's larger or smaller than 1.0. The further from 1.0, the stronger the association.

Statistical significance can be assessed by:

- Small p-values (<.05) for Chi square tests
- Small p-value for Fisher exact test
- Confidence limits for the odds ratio that don't include 1.0

The expected value of a cell is calculated by multiplying the row and column totals and dividing by the grand total. If any expected value is less than five, use the Fisher Exact test results and Exact confidence limits.

Stratified Analysis of 2-by-2 Tables

If confounding exists, associations between disease and exposure can be missed or falsely detected. A confounding factor is associated with both the disease and the exposure but may not be of direct interest.

Stratification creates separate tables for each possible confounder combination. For example, if sex is the potential confounder, you would make separate tables for males and females.

The Mantel-Haenszel statistics combine results from different strata to remove confounding. You can judge the degree of confounding by comparing the crude and weighted odds ratios. If they're identical, there was no confounding.

The confidence limits provide additional measures. If the weighted odds ratio or relative risk has confidence limits that don't include 1.0, there's a statistical association between disease and exposure without confounding by the stratifying factor.

If odds ratios vary widely between strata, interaction between the stratifying factor and risk factor may be present. You might need to use logistic regression or present the stratum-specific estimates separately.

Example 1: Bladder Cancer and Sweetener Use

A study of bladder cancer cases used randomly selected controls to explore artificial sweetener use:

Bladder Cancer

Sweetener	Yes	No
Ever Used	1293	2455
Never Used	1707	3321

1. From StatCalc, select **Tables <2 x 2, 2 x n>**
2. Enter the data from the table
3. Results appear as you enter the values

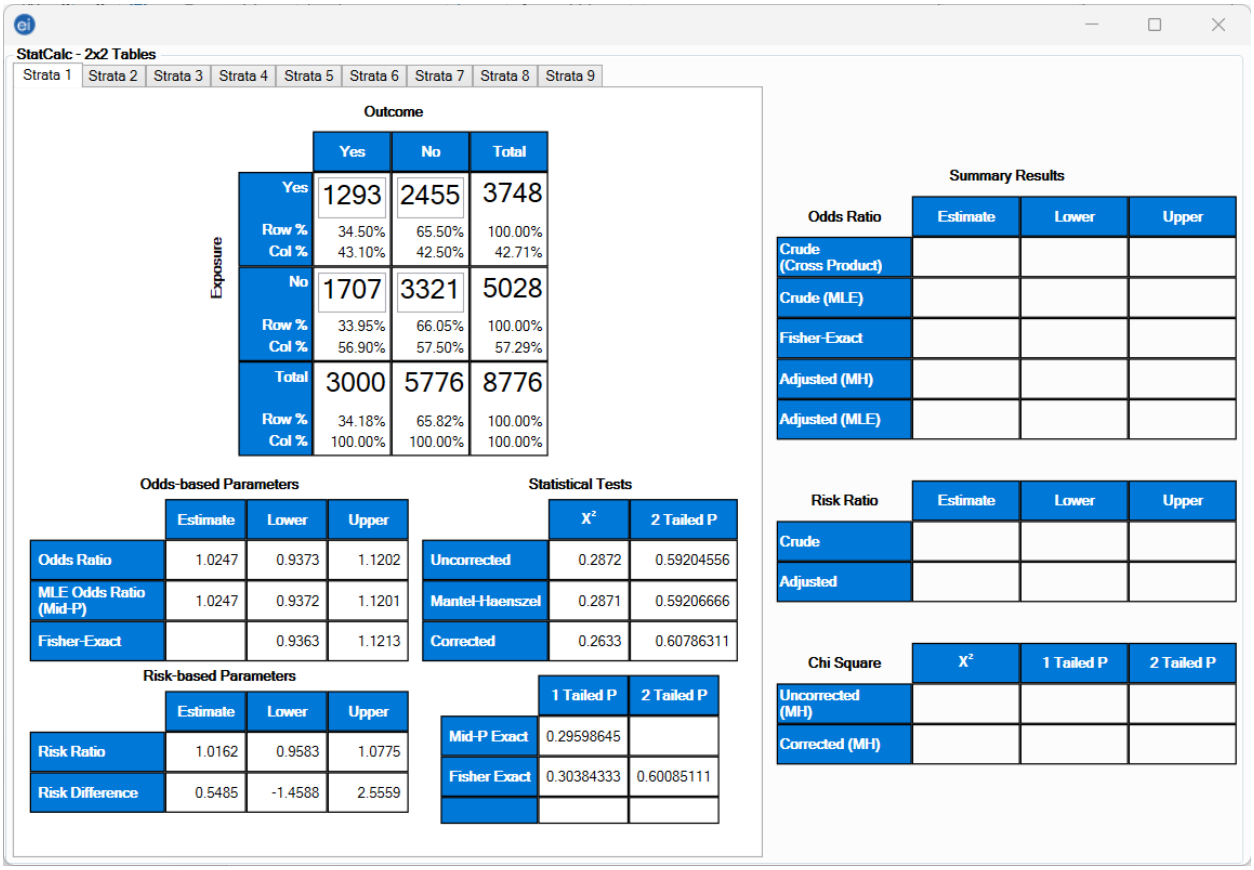


Figure 1: Results Table

The odds ratio of 1.02 and confidence limits that include 1.0 fail to show evidence of association between sweetener use and bladder cancer.

Example 2: Alcohol, Smoking, and Myocardial Infarction

This case-control study shows an apparent association between alcohol consumption and myocardial infarction (MI) with an odds ratio of 2.26:

MI

Alcohol Yes No

Yes 71 52

	MI	
No	29	48

Smoking is associated with both MI and alcohol consumption. When we stratify by smoking:

Nonsmokers:

	MI	
Alcohol	Yes	No
Yes	8	16
No	22	44

Smokers:

	MI	
Alcohol	Yes	No
Yes	63	36
No	7	4

The odds ratio for each table is 1.0, and the Mantel summary odds ratio is 1.0. Since the crude odds ratio (4.0) and Mantel summary odds ratio (1.0) differ significantly, smoking was clearly a confounding factor. There appears to be no association between alcohol and MI after accounting for smoking.

Note: that the odds ratio is the same (1.0) in both strata, showing no interaction between smoking and alcohol. The effect of alcohol on MI is the same for both smokers and nonsmokers.

To create a stratified analysis:

1. From StatCalc, select **Tables <2 x 2, 2 x n>**
2. Enter the Nonsmokers data in the first Strata tab (Strata 1)
3. Click on Strata 2
4. Enter the Smokers data
5. The Stratified Analysis Summary appears automatically

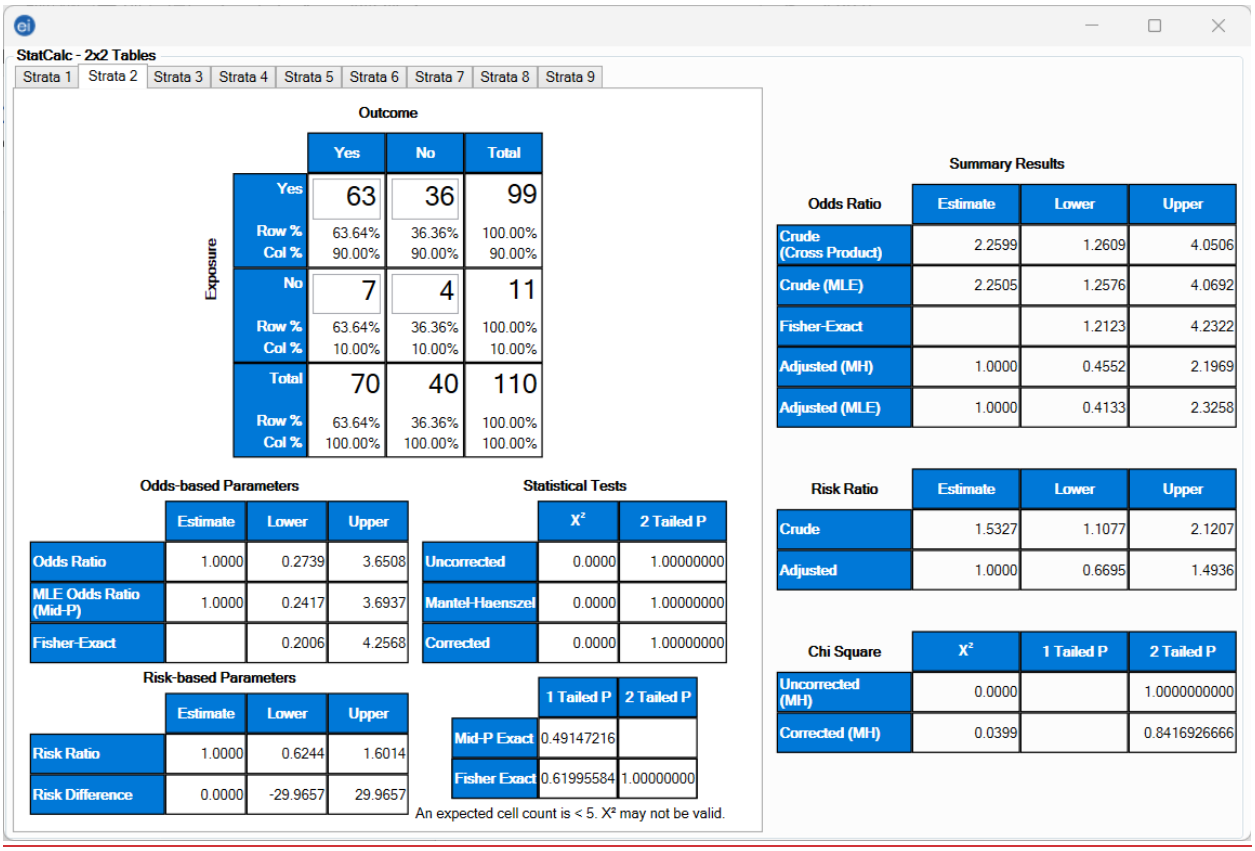


Figure 2: Multi-Strata 2x2 Results Table

- **Population Survey or Descriptive Study**
- **Assumptions**
- The sample must be simple random or representative. A systematic sample (like every fifth person on a list) is acceptable if it's representative.
- The question must have a Yes-No or other two-choice answer, resulting in a proportion of the population (those answering Yes).

Example

Suppose you want to determine if the true prevalence of HIV antibody in a population is 10%. You plan a random sample to estimate the prevalence and want 95% confidence that the true proportion falls within your calculated confidence interval.

To calculate the sample size:

1. Select **Sample size & power** from StatCalc.
2. Select **Population Survey**
3. Enter Population Size of 5,000
4. Enter Expected Frequency of 10%
5. Enter Confidence Limits as 6%

The results appear showing sample sizes for different confidence levels:

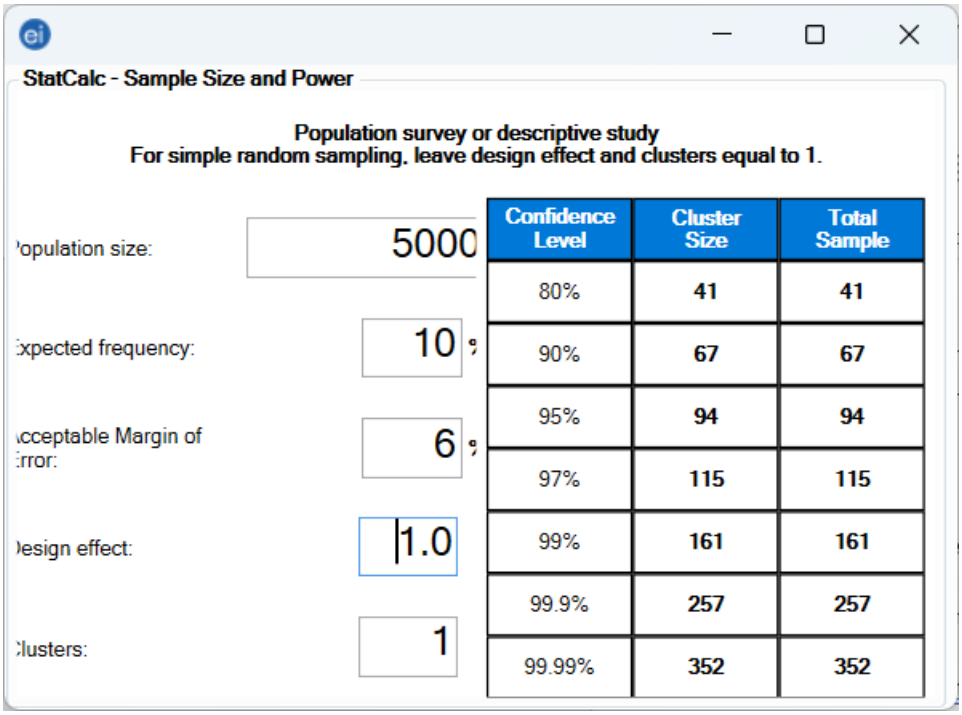


Figure 3: Population Survey or Descriptive Study Results Table

Notes

Sample size determination is only a rough guide based on assumed values for the true population proportion, sample variability, and confidence limits. Consider other factors too:

- Cost
- Number of available subjects
- Rate of non-response
- Accuracy of answers and data entry

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